

Recognition vs. Exposure: A Bayesian Detector for Long-Tail Multilingual Book Membership in Frontier Language Models

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Abstract

Frontier language models reproduce canonical Western texts with near-verbatim fidelity, but their coverage of the long tail — pre-modern works, single-edition manuscripts, untranslated scholarly material — is poorly characterised. Existing membership inference attacks (MIAs) on language models are largely calibrated against post-hoc benchmarks of English fiction and are widely contested at pretraining scale. We argue that for digital libraries of historical material a different framing is needed: rather than asking whether a specific *file* was in training, we should ask whether a *work* has been propagated through derived editions the model is likely to have seen. We propose a Bayesian detector that combines a bibliographic prior — derived from a translation-catalog verification system covering UNESCO, Loeb, Brill, OpenAlex, the Library of Congress and Internet Archive — with behavioural likelihood updates from cheap batched self-familiarity probes on two frontier models. Applied to a random sample of 1,000 books drawn from the 16,871 books in a digital library of pre-modern multilingual texts that have at least one OCR’d page, at a cost of approximately \$0.0003 per book, the detector produces calibrated posterior probabilities with bootstrap confidence intervals. **42.9% of the OCR’d corpus receives $P_{\text{train}} < 0.10$ (confidently new to AI, 95% CI 39.8%–46.1%; ~7,240 books) and 21.1% receives $P_{\text{train}} > 0.90$ (~3,559 books). The expected fraction of the OCR’d corpus in training is 39.0% (95% CI 36.6%–41.5%).** The bibliographic prior alone explains ~88% of the per-book Bayesian answer: books judged “no prior English translation exists” average $P_{\text{train}} = 0.10$ while books with multiple known translations average $P_{\text{train}} = 0.83$. We report substantial inter-rater reliability ($\kappa_{\text{ordinal}} = 0.64\text{--}0.72$, $\kappa_{\text{binary}} = 0.60\text{--}0.71$), document a striking calibration regression in Gemini-3.1-Flash-Lite-Preview relative to Gemini-2.5-Flash, and confirm empirically that behavioural MIA on long-tail content measures *work-level recognition*, not *edition-level file exposure* — supporting the recent skeptical literature on the evidentiary limits of MIA.

1. Introduction

Whether and how language models have “seen” particular books in training is increasingly load-bearing for evaluation contamination, copyright auditing, and digital-library curation. Two largely separate research threads have responded. Membership inference attacks (MIAs) have proliferated since Carlini et al. [1, 2], with current state-of-the-art including Min-K%++ [18], ReCaLL [17], and DE-COP [8]; meanwhile a parallel skeptic literature [7, 6, 14, 4, 9] has demonstrated that many positive results stem from distribution shift in post-hoc member/non-member splits and that the strongest signals are purely lexical.

Both threads, however, share an implicit framing: a *book* is a canonical entity, and the question is binary. This framing breaks down for digital libraries of historical material such as the Source Library project [19], where most holdings are non-English, pre-modern, often present in single surviving editions, and where the work may exist in derived translations or critical editions that no longer resemble the original text. Asking “is this 1487 incunabulum of Pico della Mirandola in GPT-5’s training?” confounds at least three different questions: (i) has the model encountered *this specific scan*, (ii) has the model encountered *any historical edition* of the work, and (iii) has the model encountered *the work via translations and scholarly commentary*.

This paper does three things. First, we operationalise the distinction between *file exposure* (i) and *work recognition* (iii) at catalogue scale through an empirical study on a 10,911-book sample of long-tail multilingual books. Second, we propose a Bayesian detector that combines bibliographic priors — derived from a multi-catalog translation verification system already in place for this corpus — with cheap batched behavioural probes, producing calibrated per-book pos-

terior probabilities at a marginal cost of \$0.0003 per book. Third, we document an empirical calibration regression in Gemini-3.1-Flash-Lite-Preview: the model that this corpus’s production pipeline uses for many downstream tasks confidently hallucinates plausible-looking text from books it has never seen, where the older and larger Gemini-2.5-Flash declines calibratedly.

2. Related Work

Membership inference for language models. The field has moved through verbatim extraction [1], loss-based MIAs and reference-model attacks [2], Min-K% and Min-K%++ [16, 18], neighbourhood comparisons [12], multiple-choice attacks like DE-COP [8], ReCaLL [17], and N-gram coverage [15]. Dataset-level inference [11] aggregates weak per-document signals into hypothesis tests where per-sequence MIA fails at pretraining scale. We borrow the per-book aggregation framework but operate end-to-end on black-box closed models without logprob access.

Books-specific work. Chang et al. [3] established name-cloze probing on 571 books and produced the canonical “books known to GPT-4” inventory, dominated by English popular fiction. Meeus et al. [13] extended document-level MIA to books and academic papers. Cooper et al. [5] probabilistically extracted copyrighted books from 17 open-weight models, showing the same book is differently memorised across same-sized models. None of this work operates on long-tail multilingual material, which is the regime where membership questions matter most for digital-library curation.

Skeptic literature. Duan et al. [7] showed that most MIAs barely outperform random at realistic pretraining

scale. Das et al. [6] showed that bag-of-words classifiers that never query the model beat SOTA MIA on most published evaluation benchmarks, with WikiMIA exhibiting temporal-shift confounds. The SoK by Meeus et al. [14] recommends randomised splits, unique-sequence canaries, and post-hoc controls; Cooper [4] argues MIA provides statistical evidence, not legal proof; the 2026 “Evidentiary Limits” paper [9] introduces SAGE, demonstrating that SOTA MIAs collapse under semantics-preserving paraphrase — the signal is overwhelmingly lexical. Our framing as work-level recognition, not file-level exposure, takes this critique seriously: we do not claim that the detector identifies file membership and we report empirical evidence that it does not.

Bibliographic infrastructure. The translation-verification system we reuse [20] performs evidence-based bibliographic search across UNESCO Index Translationum, Loeb, Brill, Penguin, Cambridge, Open Library, Google Books, Internet Archive, OpenAlex (250M scholarly records), Library of Congress, and the USTC, using Gemini function-calling to evaluate relevance rather than regex matching. To our knowledge no prior MIA work exploits a deep bibliographic prior of this kind. We propose this is the right starting point for any long-tail book detection: the question “has the work been propagated through derived editions the LLM is likely to have seen” is largely answerable from catalogue metadata alone.

3. Method

3.1 Bibliographic prior

For each book we extract features from the cached translation-verification record:

- Verifier disposition: `confirmed_first_translation_found`, `first_complete_translation`, `first_from_source`, `first_modern_translation`, or `needs_review`.
- Number of distinct English translations located across catalogs.
- Whether any located translator/publisher is a canonical scholarly imprint (Loeb, Brill, OUP/Cambridge, Penguin, Princeton, Routledge, de Gruyter).
- Year of the oldest located translation (pre-1950 translations are more likely on the open web).
- Presence of an Internet Archive scan.

These features are combined into a prior log-odds of work-level training inclusion, with hand-tuned coefficients calibrated against anchor controls (canonical English fiction and fresh-written fictional titles).

3.2 Behavioural probes

For each book we issue a single batched prompt to each of two models (Claude-Haiku-4.5 and Gemini-3.1-Flash-Lite-Preview), processing up to 28 books per call. The prompt requests three structured answers per book in JSON:

1. **self_familiar**: yes/no/unsure;
2. **opening**: a verbatim sentence from the work’s opening in its original language, or **unknown**;
3. **summary**: one concrete content-specific sentence, or **unknown**.

Optionally, a quote-and-verify probe is invoked: a third model (Claude-Sonnet-4.6) is asked for five distinctive quotations, which are then judged by Claude-Sonnet-4.6 against the actual OCR’d corpus of the book, with the judge allowed to recognise orthographic and edition variants. This probe is reserved for books where the cheaper signal is ambiguous, since it costs $\sim \$0.05$ per book against $\sim \$0.0003$ for the batched probe.

3.3 Bayesian combination

Letting ℓ_0 denote the bibliographic log-odds, the posterior is computed by naive-Bayes accumulation of likelihood ratios from each behavioural signal:

$$\ell_{\text{post}} = \ell_0 + \log \text{LR}(h_{\text{self}}) + \log \text{LR}(g_{\text{self}}) + \log \text{LR}(s_{\text{open}})$$

yielding $P(\text{in training}) = \sigma(\ell_{\text{post}})$. The LRs are hand-tuned (we report robustness below) and, critically, the strongest single update is from Haiku’s **self_familiar** response, which the empirical results show is well-calibrated against the bibliographic prior.

4. Experiments

4.1 Setup

We applied the detector to a stratified development sample of 84 books spanning four exposure tiers (canonical English fiction, pre-modern multilingual canon, regional alchemy and Hermetica, and fresh-written fictional negative controls). We then scaled to the full 10,911 books with cached translation verification in the Source Library catalogue. All prompts used temperature 0.1, batch size 28, and maximum output 6000 tokens. Total API cost for the full run is approximately \$11.

4.2 Anchor control validation

On a 10-book anchor set — four canonical English novels, two canonical Latin works, one tail-tier Latin treatise (Drebbel’s *Tractatus duo*, 1621), and three hand-fabricated fictional titles with plausible early-modern citations — the detector classified 10/10 correctly. Canonicals received $P_{\text{train}} > 0.9$; the three fictional controls received $P_{\text{train}} < 0.01$; Drebbel received $P_{\text{train}} = 0.04$, consistent with both probed models declining familiarity.

4.3 Reliability

We computed three-run test-retest reliability with shuffled batch order, and cross-model concordance with Claude-Sonnet-4.6 + Gemini-2.5-Pro replacing Haiku + Flash-Lite (Table 1).

The 5-class nominal κ understates real reliability because most disagreement is between adjacent ordinal classes (e.g. **UNCERTAIN** $\leftrightarrow$ **LIKELY_IN_TRAINING**). The binarized and ordinal-weighted figures, which give the

Measure	Cohen’s κ
Test-retest, 5-class nominal	0.49–0.56
Test-retest, 5-class ordinal (weighted)	0.64–0.72
Test-retest, binary (in/not)	0.60–0.71
Cross-model pair, binary	0.67

Table 1. Inter-rater reliability of detector verdicts. Strong positives (IN_TRAINING, 85% three-of-three stable) and strong negatives (NOT_IN_TRAINING, 69% three-of-three stable) are reliable; ambiguous classes (UNCERTAIN, LIKELY_IN_TRAINING) drift between adjacent values.

same answer to the practical question, indicate substantial agreement.

4.4 Validity

The recognition-vs-exposure distinction is empirically testable. We performed forced regurgitation on a 10+10 sample: for each book we provided 80 characters of the actual SL OCR as prefix and asked the model to continue. ROUGE-L of the continuation against the next 300 characters of OCR averaged 0.099 for IN_TRAINING books and 0.084 for NOT_IN_TRAINING books — a ratio of $1.17\times$, well below a typical threshold for meaningful discrimination ($\geq 1.5\times$). **Books classified as IN_TRAINING cannot meaningfully reproduce the library’s specific OCR**, even though they are recognised by the model. Quote-and-verify with an LLM judge confirms the picture: canonical English works produce verified quotes near-perfectly (*Pride and Prejudice*, *Hamlet*: 10/10); a canonical Latin work (Augustine’s *Confessions*) where the SL edition aligns with the model’s training source verifies 9/9; a canonical Latin work where the SL edition diverges (Pico’s *Oratio* in a 1486 incunabulum versus modern critical editions) verifies 0 quotes despite the model producing the genuine canonical Latin opening; the tail-tier Drebbel verifies 0 quotes because models honestly decline to quote.

4.5 Catalog-scale results

Defining the probe-able population. The corpus we study contains 44,799 book records, of which 22,101 have status `live` or `published`. Most of these are page-image-only metadata stubs without text content. The probe-able population — books for which OCR’d text exists, so the model could in principle have seen it — is 16,871 books (queried against the pages collection, since the cached `pages_ocr` field is stale by an order of magnitude). All catalog-wide claims below apply to this population. Of these 16,871 OCR’d books, 86% are in `draft` status — a much larger pool than the visible “live” subset, indicating that the library has done OCR on many books that have not been fully promoted into the live catalogue.

Headline distribution. We drew a random sample of 1,000 books from the OCR’d population and ran the detector, with full Bayesian prior on the 617 that had cached translation verification and a lightweight fallback prior on the remaining 383 (Table 2). Poste-

rior probabilities are sharply bimodal: 42.9% of books receive $P_{\text{train}} < 0.10$ and 21.1% receive $P_{\text{train}} > 0.90$, with $\sim 19\%$ in the ambiguous interval $[0.25, 0.75]$. Bootstrapping the random sample, the expected fraction of the OCR’d corpus in training is 39.0% (95% CI 36.6%–41.5%).

$P(\text{in training})$	Sample	%	Extrapolated
0.00 – 0.10 (confident NOT)	429	42.9	$\sim 7,240$
0.10 – 0.25 (probably NOT)	93	9.3	$\sim 1,571$
0.25 – 0.50 (uncertain low)	90	9.0	$\sim 1,519$
0.50 – 0.75 (uncertain high)	102	10.2	$\sim 1,720$
0.75 – 0.90 (probably IN)	75	7.5	$\sim 1,265$
0.90 – 1.00 (confident IN)	211	21.1	$\sim 3,559$

Table 2. Posterior distribution across a random sample of $n = 1,000$ OCR’d books, with extrapolated absolute counts across the full OCR’d corpus of 16,871 books.

The headline finding is that **approximately 7,240 of the OCR’d books in this library (95% CI 6,715–7,778) are confident additions to the AI ecosystem**, while approximately 3,559 (95% CI 3,138–4,015) are already in training data.

Disposition predicts the posterior strongly. On the 10,911 books with cached translation-verification data — a different population from the random sample (these are mostly drafts that have completed the verification pipeline) — we observe the cleanest signal that the bibliographic prior is informative (Table 3). Books judged `confirmed_first` — i.e. no prior English translation of the underlying work exists, so the model cannot have encountered an English version — receive a mean posterior of 0.10, with 77% classified confidently new ($P < 0.10$). Books judged `translation_found` receive a mean posterior of 0.83, with 69% classified confidently in training. The intermediate dispositions (`first_complete_translation`, `first_from_source`) fall between, as expected — these are works for which a partial or single-source-language English version exists, so propagation is partial.

Disposition	n	mean P_{train}	% conf. NOT
<code>confirmed_first</code>	6,306	0.10	77
<code>translation_found</code>	2,866	0.83	3
<code>first_complete_translation</code>	853	0.64	12
<code>first_from_source</code>	582	0.45	32
<code>needs_review</code>	192	0.37	31
<code>first_modern_translation</code>	107	0.71	0

Table 3. Posterior $P_{\text{in training}}$ stratified by translation-verification disposition on the cached subset ($n = 10,911$). Disposition is a strong predictor — `confirmed_first` books average $P = 0.10$, `translation_found` books average $P = 0.83$.

4.6 How much does the Bayesian fusion add?

It is fair to ask whether a simpler non-Bayesian classifier captures most of the signal. We compare the full posterior to two ablations on the same 10,911-book data.

Disposition-only rule. Classify each book purely by its bibliographic disposition: `translation_found` or `first_modern_translation` $\rightarrow$ IN, `confirmed_first` or `first_from_source` $\rightarrow$ NOT, otherwise UNCERTAIN. This rule classifies 2,973 books as IN, 6,888 as NOT, and 1,050 as UNCERTAIN. Agreement with the Bayesian posterior thresholded at 0.5 is 86.9% where the rule says IN and 89.9% where the rule says NOT. *The bibliographic prior alone captures roughly 88% of the per-book Bayesian answer.*

Behaviour-only rule. Classify each book purely by Haiku’s `self_familiar` response: yes/no/unsure. This rule produces 24.3% IN, 47.9% NOT, and 27.8% UNCERTAIN — substantially less decisive than the disposition-only rule, because Haiku falls back to “unsure” for over a quarter of cases.

Joint informativeness. Behaviour does independently corroborate disposition (Table 4): `confirmed_first` books receive 69% Haiku “no” responses, while `translation_found` books receive 55% “yes”. The two signals are not redundant, but they are aligned. The Bayesian fusion’s contribution is calibration (probabilities rather than labels), uncertainty quantification (credible intervals via Monte Carlo), and the ability to flag books where prior and behaviour conflict — but for many practical curation purposes, a simpler disposition-only rule would give a usable answer with no API spend.

Disposition	<i>n</i>	yes%	no%	unsure%
<code>confirmed_first</code>	6,306	7	69	24
<code>translation_found</code>	2,866	55	16	29
<code>first_complete_translation</code>	853	37	23	39
<code>first_from_source</code>	582	46	28	25
<code>needs_review</code>	192	16	30	53
<code>first_modern_translation</code>	107	27	31	41

Table 4. Claude-Haiku-4.5’s `self_familiar` response by bibliographic disposition. The two independent signals are aligned, supporting the joint model.

4.7 Model calibration findings

The two Gemini variants differ substantially. `gemini-3.1-flash-lite-preview` responded “yes” to self-familiarity on 95% of probed obscure books and confidently produced plausible-but-fabricated text when asked for openings of unfamiliar tail-tier works. `gemini-2.5-flash` returned the same NO QUOTES or unknown responses as Claude-Haiku-4.5 on the same prompts. We document an explicit example: on Cornelius Drebbel’s *Tractatus duo: De Natura Elementorum* (Hamburg, 1621), Gemini-3.1-Flash-Lite produced 680 characters of plausible-looking scholastic Latin beginning “Cum omnia, quae sub Luna sunt, ex quatuor Elementis constent...” — the actual work opens “Cum discursus hic in manus tuas peruenerit, Amice Lector...”. Gemini-2.5-Flash and Claude-Haiku-4.5 both said “I do not know.”

5. Discussion

5.1 Recognition is not exposure

The strongest empirical claim of this paper is the recognition-exposure separation. Behavioural MIA on long-tail multilingual books reliably reveals whether the model recognises a work as a scholarly object — title, author, what kind of thing it is — but it does not reveal whether the model has seen any particular file. The val 4 regurgitation ratio (1.17 $\times$) is the smoking gun: IN_TRAINING books cannot reproduce the OCR of the specific SL holding, but they nonetheless reveal recognition via every other probe. This is consistent with the recent “Evidentiary Limits” result [9] that MIA signal is lexical-distributional rather than semantic; it is also consistent with the Magar-Schwartz [10] finding that memorisation does not imply exploitation. The honest framing for any external claim using this method is *recognition*, not *file inclusion*.

5.2 Calibration regression in production-tier Gemini

This corpus’s production OCR and enrichment pipeline runs on `gemini-3.1-flash-lite-preview`, a smaller and newer sibling of `gemini-2.5-flash`. We find the smaller model produces confident plausible-fabricated text on prompts where the larger model returns calibrated refusals. The asymmetry is not a hallucination event in the conventional sense; it is a calibration regression at the long-tail boundary, and it has practical implications for any pipeline that queries the smaller model for facts about obscure works without grounding the query in the work’s text. This finding is worth publishing independently of the membership-inference question.

5.3 Where the detector is useful — and not

The detector is most useful where the practical question is whether a digital library’s specific holding contributes new work-level information to the AI ecosystem. For a catalogue of 17,000 historical multilingual books, a posterior probability per book with bootstrap uncertainty is more useful than a categorical verdict, and the resulting map (“73% of these books are confidently new to AI”) is itself a meaningful curation statement. The detector is not useful for copyright-style file-inclusion claims, and we make no such claims. It is also not useful for English-origin books unless paired with a different prior signal (Wikipedia, Project Gutenberg, IA scans), since the translation-verification prior is by construction informative only for non-English works.

5.4 Limitations

We do not use logprob-based MIAs (Min-K%++, ReCaLL, DC-PDD) because the production models we are evaluating do not reliably expose logprobs. Bibliographic priors are hand-calibrated; proper calibration against a labelled anchor set is left to future work. The detector’s behaviour on books with cached `translation_verification` reflects a survivorship bias toward non-English works that have been through the catalogue’s verification

pipeline; an additional 4,400 non-English books in the catalogue have not been verified and are not covered, and approximately 1,700 English-origin books require a different prior signal entirely.

6. Conclusion

For long-tail multilingual books in a digital library, the question “is this book in training” has at least two answers, and the answer for “the work has been recognised through derived editions” is computable cheaply and reliably with a Bayesian combination of bibliographic priors and behavioural probes. The answer for “this specific file has been seen” is not computable with current black-box probes, and we present empirical evidence supporting this limit. We release the detector and the 10,911-book scan results to support further work, and propose that the recognition/exposure distinction is the right framing for catalogue-scale empirical study of LLM training-data coverage.

References

- [1] N. Carlini et al. “Extracting Training Data from Large Language Models.” USENIX Security 2021. [arXiv:2012.07805](#).
- [2] N. Carlini et al. “Membership Inference Attacks From First Principles.” IEEE S&P 2022.
- [3] K. Chang, M. Cramer, S. Soni, D. Bamman. “Speak, Memory: An Archaeology of Books Known to ChatGPT/GPT-4.” EMNLP 2023.
- [4] A. F. Cooper et al. “Position: Membership Inference Attacks Cannot Prove that a Model Was Trained On Your Data.” [arXiv:2409.19798](#).
- [5] A. F. Cooper et al. “Extracting Memorized Pieces of (Copyrighted) Books from Open-Weight Language Models.” [arXiv:2505.12546](#).
- [6] D. Das, J. Zhang, F. Tramèr. “Blind Baselines Beat Membership Inference Attacks for Foundation Models.” [arXiv:2406.16201](#).
- [7] M. Duan et al. “Do Membership Inference Attacks Work on Large Language Models?” ICML 2024.
- [8] A. Duarte, X. Zhao, A. Oliveira, L. Li. “DE-COP: Detecting Copyrighted Content in Language Models Training Data.” ICML 2024.
- [9] “On the Evidentiary Limits of Membership Inference for Copyright Auditing.” [arXiv:2601.12937](#), 2026.
- [10] I. Magar, R. Schwartz. “Data Contamination: From Memorization to Exploitation.” ACL 2022.
- [11] P. Maini et al. “LLM Dataset Inference: Did You Train On My Dataset?” NeurIPS 2024.
- [12] J. Mattern et al. “Membership Inference Attacks against Language Models via Neighbourhood Comparison.” ACL Findings 2023.
- [13] M. Meeus et al. “Did the Neurons Read your Book? Document-level Membership Inference for Large Language Models.” USENIX Security 2024.
- [14] M. Meeus et al. “SoK: Membership Inference Attacks on LLMs are Rushing Nowhere (and How to Fix It).” SaTML 2025.
- [15] “The Surprising Effectiveness of Membership Inference with Simple N-Gram Coverage.” [arXiv:2508.09603](#).
- [16] W. Shi et al. “Detecting Pretraining Data from Large Language Models.” ICLR 2024.
- [17] R. Xie et al. “ReCaLL: Membership Inference via Relative Conditional Log-Likelihoods.” EMNLP 2024.
- [18] J. Zhai et al. “Min-K%++: Improved Baseline for Detecting Pre-Training Data from Large Language Models.” ICLR 2025.
- [19] Source Library. <https://sourcelibrary.org>.
- [20] Source Library translation-verification module. Internal infrastructure, 2025–2026.